INCH-POUND

MIL-S-52302D
18 June 1992
SUPERSEDING
MIL-S-52302C
13 October 1983

MILITARY SPECIFICATION

STRUT, BRACING, TRESTLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trestle bracing strut to be used with the trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-889	-	Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13207E9733	-	Strut, Bracing, Trestle.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953	-	Strapping, Flat Steel and Seals.
D 4675	-	Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. (NMFTA)

National Motor Freight Classification Rules

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(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The strut shall be as shown on top assembly TA13207E9733 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the strut. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be new and unused.

3.3.1 Material deterioration and control. The strut shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

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3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Identification marking. The strut shall be metal die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.5 Treatment and painting. The portions of the item normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green, color No. 34094. If required the item shall then be overcoated in accordance with the Government furnished camouflage patterns with the colors specified in the camouflage patterns

3.6 Government furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.7).

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

3.7 Stenciling. The nomenclature, national stock number, weight, and cube shall be stenciled on each strut. The letters "NET WT" shall precede the numerals in the weight and letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 0.75 inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship. Holes shall be drilled through both walls of the tubing in one operation through a metal template, jig, or fixture so as to be aligned true, square, and normal to the tubing. All burrs, rough edges, and edges of holes shall be ground smooth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform

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any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.5).

4.3 First article inspection.

4.3.1 Examination. The first article strut shall be examined for the defects specified in 4.4.2. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

Major

101. Any part not in accordance with the QAP requirements as shown on the drawings (see 3.1).
102 Materials not as specified (see 3.3).

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103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment (see 3.3.1)..
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Contractor does not have documentation available for identification of material, material finishes or treatments (see 3.3.3).
106. Workmanship not as specified (see 3.8).

Minor

201. Dimensions other than those identified by QAP data not as specified (see 3.1.1).
202. Identification marking incorrect, missing, or illegible (see 3.4).
203. Treatment and painting not as specified (see 3.5).
204. Stenciling not as specified (see 3.7)..

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. One or more defects shall be cause for rejection.

107. Bundles not as specified for level A (see 5.1.1).
108. Strapping not as specified for level A (see 5.1.1).
109. Strapping size and placement not as specified for level A (see 5.1.1).
110. Level C packing not as specified (see 5.1.2).
111. Marking missing, illegible, incorrect for level A or C (see 5.2).

5. PACKAGING

5.1 Packing (see 6.6). Packing shall be level A or C as specified (see 6.2).

5.1.1 Level A. The struts shall be bundled together in quantities of six. Each bundle shall be secured with not less than five flat steel straps conforming to ASTM D 3953, type 1 or 2, zinc-coated, size 3/4 by 0.020 and ASTM D 4675. The straps shall be placed approximately 18 inches from each end of the bundle, and the intermediate straps shall be equally spaced between the end straps.

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5.1.2 Level C. The struts shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking (see 6.6). In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The strut is intended for use in floating and fixed bridges for bracing trestles to obtain lateral and longitudinal stability.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and the number of units required (see 3.2).
d. When camouflage patterns will be provided (see 3.5 and 6.7).
e. Level of packing required (see 5.1).
f. Any special marking requirement (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing

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requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definition.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Levels of preservation. Levels of preservation have not been included as the struts do not require additional preservation. For purposes of preservation/ packing level marking, the preservation shall be designated as level A.

6.7 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.6.

6.8 Subject term (key word) listing.

Strut
Trestle assembly
Trestle bracing strut

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
Navy - MC
DLA - CS

Preparing activity:
Army - ME

Project 5420-0203

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
   The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-52302D	2. DOCUMENT DATE (YYMMDD) 92/06/18

3. DOCUMENT TITLE
Strut, Bracing, Trestle

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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